

Lantern OS Whitepaper v0.1

Generated: 2026-05-26.

Repo: `https://github.com/alex-place/lantern-os`

Status: draft canonical whitepaper for local-first Lantern OS, COMET LEAP, RAG dollhouse, dual boot, local wallet, and One World Leader app convergence.

Abstract

Lantern OS is a local-first operating spine for turning scattered knowledge, repos, PDFs, images, device plans, and cash offers into one evidence-backed working system.

The thesis is simple:

knowledge -> evidence -> RAG -> packet -> invoice -> cash -> better local intelligence

Lantern OS does not claim to contain all knowledge. It creates a repeatable path for compressing verified knowledge into useful decisions, printable reports, learning packets, local AI workflows, and paid services.

Problem

Modern AI work usually fails in four places:

- 1. Knowledge is scattered across repos, PDFs, chats, devices, browser tabs, and old project folders.
- 2. Confidence is guessed instead of tracked.
- 3. Product ideas expand faster than cash validation.
- 4. Local hardware, privacy, boot safety, and family/school needs are treated as afterthoughts.

Lantern OS fixes this by making every claim, artifact, offer, wallet event, and device boundary visible.

System

Lantern OS has seven control surfaces:

Surface	Purpose	Current Path
RAG Dollhouse	single flat memory and copied assets	`skills/lantern-rag-dollhouse/`
COMET LEAP	agile convergence and report method	`skills/comet-leap-agile/`
Bayesian World Model	priors, evidence, posteriors, decisions	`data/world-model/belief-ledger.jsonl`
Product Atlas	knowledge-to-cash product map	`reports/LANTERN-PRODUCT-UNIVERSE-ATLAS.md`
Local Wallet	invoice and factual cash ledger	`data/wallet/local-cash-wallet.json`
Dual Boot / Local OS	Windows/NixOS readiness and boundaries	`dual-boot/`
One World Leader	education, leadership, commons, packets	`skills/one-world-leader-app/`

RAG Dollhouse

The RAG dollhouse is the memory body. It is not a vague idea; it is a flat file plus literal assets.

Canonical flat file:

```
skills/lantern-rag-dollhouse/references/LANTERN-OS-RAG-DOLLHOUSE.flat.md
```

Asset bundle:

```
skills/lantern-rag-dollhouse/assets/
```

The current bundle includes COMET LEAP PDFs, 30-day art images, chart images, the Bitcoin PDF, and the Ethereum whitepaper PDF. Every copied asset is hashed in `skills/lantern-rag-dollhouse/assets/ASSET-MANIFEST.sha256`.

Local Wallet

The Lantern wallet is not a bank account, crypto wallet, Stripe account, or fake balance. It is a local operating ledger for the 11-day cash sprint.

```
clearedCashUsd: 0<br/>pendingInvoiceUsd: 199<br/>firstInvoice: INV-COMET-LEAP-RAG-001<br/>firstOffer: Local RAG / Repo Cleanup Sprint
```

Rules:

- existing offers only;
- do not count payment until funds clear;
- record draft, sent, promised, cleared, refund, and objection as separate events;
- keep private payment and customer secrets out of Git.

Governance

Lantern OS uses offchain, evidence-first governance inspired by open protocol practice: proposals are written down, reviewed, validated, iterated, and only promoted when the evidence and operator boundary allow it.

Ethereum governance is a useful comparison because protocol changes require public discussion, client implementation, coordination, testing, and broad stakeholder support rather than a single owner forcing changes. Lantern OS uses the same spirit locally:

Ethereum Governance Idea	Lantern OS Adaptation
EIPs propose protocol changes	manifests and reports propose repo changes
all-core-dev style review	orchestrator and operator review
testnets before mainnet	local validation before v1.0.0
hard fork risk	branch/repo/surface split risk
offchain consensus	evidence-backed operator approval

This keeps the project fast without pretending every idea is automatically production-ready.

Dual Boot

Current PC status:

```
readyForPrep: true<br/>readyForInstall: false<br/>failures: 0<br/>primary blocker: 0.0 GB unallocated install space
```

The system is dual-boot-capable. It does not show installer-ready because Windows has no unallocated install space yet. D: has about 1.6 TB free and is the obvious shrink candidate.

Hard boundary: Lantern OS may inspect, document, and validate. It must not resize partitions, mutate BCD, change firmware boot order, format disks, or install an OS unattended.

One World Leader App

The first app slice is:

```
One World Leader -> Product Atlas -> 11-Day Cash Sprint -> Local Wallet -> One paid packet
```

The app is not a claim of political authority. It is a leadership and learning workbench that compresses world knowledge into:

- understandable maps;
- evidence-backed decisions;
- art, music, math, science, and game learning packets;
- public-domain and Creative Commons media indexes;
- ethical AI/RAG workflows;
- cashable productized services.

11-Day Cash Sprint

The cash sprint refuses abstract scale until cash conversations happen.

Existing offers:

Offer	Price	First Buyer
COMET LEAP Founder Report Pack	`\$99-\$299`	founder / indie builder
Local RAG / Repo Cleanup Sprint	`\$199-\$999`	builder / consultant
Windows / Lantern Setup Session	`\$99-\$299`	local user / family
Parent / Homeschool Creative Learning Packet	`\$49-\$149`	parent / teacher
Small-Business AI Cleanup	`\$199-\$750`	SMB owner

For the sprint, count only calls booked, invoices sent, cash collected, paid pilots delivered, and objections recorded.

Token And Compute Policy

Offline/local/server-farm Foundry tokens are unmetered internal capacity. They are not "Lite", not rated per token, and not billed as cloud burn.

Cloud/API escalation remains metered and must use current provider pricing.

Crypto whitepapers are architecture anchors for ledger discipline and state-machine thinking, not authorization for speculative token issuance.

Evidence Anchors

Evidence	Path	Use
Bitcoin PDF	`skills/lantern-rag-dollhouse/assets/pdfs/bitcoin.pdf`	peer-to-peer cash and ledger discipline
Ethereum whitepaper PDF	`skills/lantern-rag-dollhouse/assets/pdfs/Ethereum_Whitepaper_-_Buterin_2014.pdf`	programmable ledger and state-machine design
Ethereum governance page excerpt	operator-provided, 2026-05-26	offchain governance analogy
COMET LEAP cash sprint	`reports/COMET-LEAP-11-DAY-CASH-SPRINT.md`	manual-first money loop
Product universe atlas	`reports/LANTERN-PRODUCT-UNIVERSE-ATLAS.md`	knowledge compression ladder

Release Rule

Lantern OS is not v1.0.0 until the operator explicitly approves it.

Promotion requires:

- clean convergence loop;
- validated RAG flat file and asset hashes;
- wallet ledger with factual events;
- dual-boot prep or held boundary documented;
- at least one cash-loop result or objection recorded;
- v1 readiness gates reviewed.

One Sentence

Lantern OS is a local-first knowledge and action operating system: it turns world evidence into useful packets, useful packets into cash, and cash back into better local intelligence.